

Christian Garry

MSc Scientific Computing | Probability · Statistics · Optimisation | C++/Python

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Education

MSc Scientific Computing & Data Analysis (AI for Engineering) Durham University	Durham, United Kingdom Sep 2025 – Exp. Sep 2026
• Current average: Distinction (89%). • Focus: probability, statistics, optimisation, numerical methods, Bayesian and regression models, HPC/GPU.	
MEng Electronic Engineering Durham University	Durham, United Kingdom Sep 2020 – Jun 2024
• Upper Second-Class Honours (2:1). • Dissertation: <i>Silicon Carbide JFET CPU</i> — simulation-driven CPU design with instruction-level verification.	

Experience

Industrial Tutor Durham University, Department of Engineering	Durham, United Kingdom Sep 2025 – Present
• Led weekly design tutorials, mentoring teams on feasibility-driven decision-making under cost, risk, and performance constraints. • Assessed and provided structured feedback on technical feasibility, assumptions, and justification quality.	
Graduate Communications Engineer Siemens PLC	Hebburn, United Kingdom Sep 2024 – Present
• Implemented and adapted C/C++ communication components (TCP/IP, serial) to support data exchange between simulated systems and PC-based engineering tools. • Developed a Python-based RAG pipeline to structure and query internal technical documentation and codebases using a retrieval-augmented workflow (local LLMs, vector search) reducing engineer time to answer by >90%.	

Projects & Research

Silicon Carbide JFET CPU Master's Dissertation	Durham, United Kingdom Oct 2023 – Apr 2024
• Independently designed and simulated a 4-bit CPU using SiC JFET logic in LTspice, targeting computation in extreme temperature and radiation environments. • Constructed a full verification toolchain (C++ compiler, assembler, Python automation) and validated instruction-level behaviour via emulator parity checks and waveform-based state reconstruction. • Investigated architectural trade-offs, race conditions, and storage constraints, explicitly documenting assumptions, limitations, and scalability bottlenecks.	

Certifications

- MITx 15.455x – Mathematical Methods for Quantitative Finance (In Progress)
- Columbia University – Introduction to Financial Engineering and Risk Management

Leadership, Activities & Interests

Research Interests: Time-series modelling, signal validation, regime dependence in financial systems.
Leadership: Bishops Diocesan College Fencing Team Captain; Durham Fresher Representative (2022–23).
Activities: Durham University Fencing Team; Boxing (Student Fight Night).
Achievements: South Africa U17 Fencing Champion & Weldon le Huray Scholar; President's Award (Gold).

Key Skills

Probability & Statistics; Numerical Methods; Optimisation; Monte Carlo Simulation; Python; C++;
High-Performance Computing; Linear Algebra